One Source to all destination

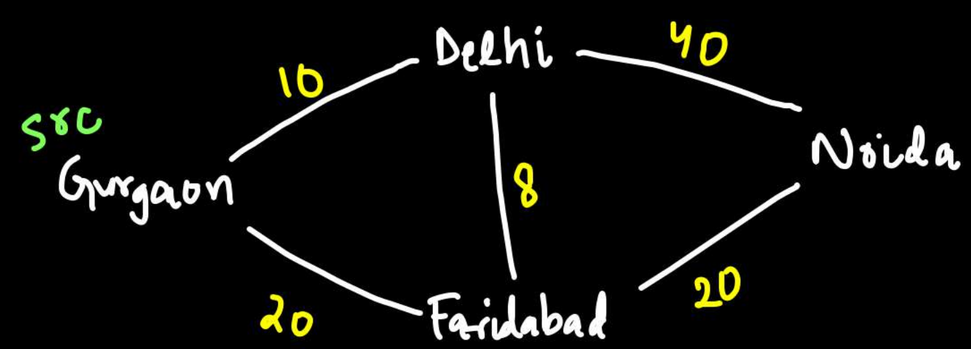
modified

Ques: Dijkstra's Algorithm (BFS) → (PQ)

Shortest Dist

Shortest Time

'edge weight'



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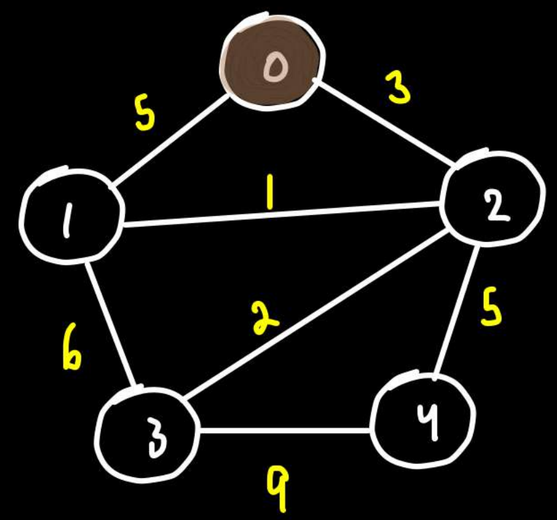
Algo



0 : (1,5), (2,3)

1 : (0,5), (2,1), (3,6)

2 : (0,3), (1,1), (3,2), (4,5)

	0	1	2	3	4
ans	0	4	3	5	8

(3, 5) → 1, 11 ∝ (6) ; 2, 7 ∝ (2) ; 4, 14 ∝ (9)

(4, 8) → 2, 13 ∝ (5) ; 3, 17 ∝ (9)

(0, 0) → (1, 5) (5) ; (2, 3) (3)

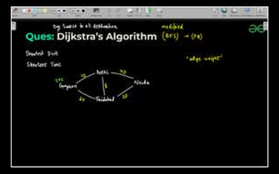
(2, 3) → 0, 6 ∝ (3) ; 1, 4 ✓ (1) ; 3, 5 ✓ (2) ; 4, 8 ✓ (5)

(1, 4) → 0, 9 ∝ (5) ; 2, 5 ∝ (1) ; 3, 10 ∝ (6)

(1, 5) ∝

pq < Node, dist >

(minheap)



Document has just been saved...

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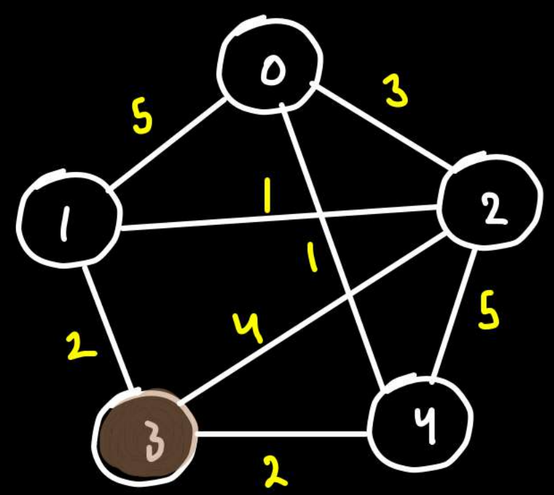
Algo



	0	1	2	3	4
ans	~~7~~ 3	2	~~4~~ 3	0	2

(2,3) → 0, 6 α; 1, 4 α; 3, 7 α; 4, 8 α

(2, 4) → α (0, 7) → α

(3, 0) → 1, 2 ✓; 4, 2 ✓; 2, 4 ✓

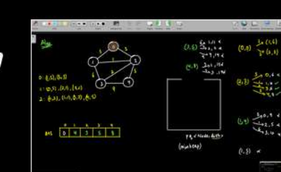
(1, 2) → 0, 7 ✓; 2, 3 ✓; 3, 4 α

(4, 2) → 0, 3 ✓; 2, 7 α; 3, 4 α

(0, 3) → 1, 8 α; 2, 6 α; 4,

pq <Node, dist>

(minheap)



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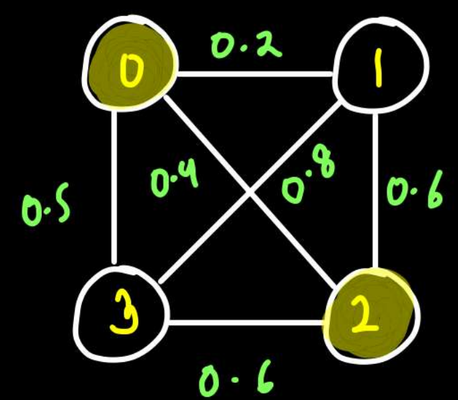
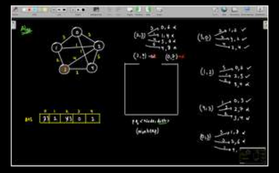
Ques: Path with maximum probability



$0 \to 1 \to 2 \qquad 0.2 \times 0.2 = 0.12$

$0 \to 2 \qquad \boxed{0.4}$

$0 \to 3 \to 2 \qquad 0.5 \times 0.6 = 0.3$